

YAKUTAT, AK, USA

WMO: 703610

Lat: 59.512N

Lon: 139.671W

Elev: 33

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25339

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.5	8.2	-7.5	3.7	11.2	-2.6	4.8	15.3	25.5	37.6	22.8	36.9	0.9	100	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	68.6	55.8	64.8	55.6	62.2	55.3	59.0	64.0	57.8	62.0	56.8	60.5	6.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.1	69.5	59.8	55.9	66.6	58.8	54.9	64.2	57.9	25.6	63.9	24.9	62.3	24.2	60.4	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	16.6	13.3	DB	-1.2	76.1	5.3	6.0	-5.0	80.5	-8.1	84.0	-11.0	87.3	-14.8	91.7	(4)
(5)			WB	-1.7	60.8	5.1	2.5	-5.4	62.6	-8.4	64.1	-11.3	65.5	-15.1	67.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.8	28.6	30.2	31.6	38.2	45.2	51.5	55.0	54.5	49.3	41.7	33.2	30.2
	DBStd	11.30	9.18	7.52	6.81	4.71	4.35	3.69	2.68	2.94	4.09	5.00	7.32	7.95
	HDD50	3761	662	553	571	355	160	22	1	3	59	258	504	613
	HDD65	8821	1127	973	1036	805	614	404	310	326	472	722	954	1078
	CDD50	415	0	0	0	0	11	67	156	142	38	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	34	0	0	0	0	0	17	7	10	0	0	0	0
	CDH80	5	0	0	0	0	0	2	1	2	0	0	0	0
Wind	WSAvg	5.1	5.2	5.1	5.5	5.1	5.3	4.7	4.4	4.3	5.0	5.3	5.2	5.6
Precipitation	PrecAvg	145.8	12.3	10.5	10.1	8.9	8.7	6.3	8.0	12.6	19.3	20.7	13.8	14.3
	PrecMax	250.2	31.8	32.1	37.3	19.1	18.9	18.3	18.7	28.3	48.3	48.8	42.7	35.2
	PrecMin	102.4	2.4	0.2	2.5	1.9	1.6	1.3	1.9	2.4	2.4	3.0	1.8	3.8
	PrecStd	30.7	6.5	6.0	6.1	4.2	4.3	4.1	3.9	6.8	7.6	7.9	6.9	7.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	45.8	51.5	62.9	69.9	76.5	71.9	73.7	69.1	55.9	50.0	48.3
		MCWB	43.3	40.9	42.7	48.3	51.8	57.4	58.5	57.8	54.4	46.9	47.1	44.2
	2%	DB	43.2	43.4	46.4	55.1	63.5	66.1	65.9	67.0	62.7	53.4	47.2	44.1
		MCWB	41.3	40.5	40.3	44.0	49.6	55.6	57.7	57.1	53.1	48.9	44.9	42.0
	5%	DB	41.3	41.6	43.5	50.2	57.7	61.4	63.0	63.7	59.1	51.6	45.0	42.2
		MCWB	39.7	39.5	39.1	42.2	48.6	54.2	56.8	56.8	53.0	48.1	42.6	40.3
	10%	DB	39.5	39.8	41.1	47.0	54.0	58.8	61.0	61.2	57.0	49.8	43.0	40.0
		MCWB	38.2	38.0	37.8	41.8	47.8	53.0	56.1	56.3	53.1	46.7	40.8	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	43.3	45.4	49.7	53.5	59.4	60.5	61.1	57.8	52.5	48.4	45.5
		MCDB	45.0	44.3	48.7	61.3	64.6	69.7	66.3	67.3	60.5	53.3	49.3	47.7
	2%	WB	41.9	41.4	42.4	45.9	51.3	56.8	58.8	59.2	56.2	50.8	45.8	42.4
		MCDB	42.9	42.6	45.0	52.4	58.7	63.3	63.7	63.6	58.9	51.9	47.1	43.9
	5%	WB	40.0	39.8	40.2	43.8	50.1	55.1	57.6	58.0	54.8	48.9	43.4	40.5
		MCDB	41.0	41.1	42.2	48.2	56.2	60.4	61.7	62.0	57.7	50.5	44.7	41.8
	10%	WB	38.3	38.2	38.4	42.3	48.5	53.7	56.7	56.9	53.4	47.0	41.2	38.6
		MCDB	39.3	39.6	40.5	46.1	53.2	57.4	60.3	60.5	56.2	48.9	42.7	39.8
Mean Daily Temperature Range	5% DB	MDBR	10.0	12.4	14.2	14.6	13.9	11.6	10.2	12.2	13.3	12.3	10.9	9.5
		MCDBR	8.3	11.7	16.9	23.2	25.6	20.1	16.7	20.3	20.7	15.4	11.4	9.5
		MCWBR	7.2	9.1	11.9	14.0	14.2	11.2	9.9	12.4	13.3	11.2	9.6	8.0
	5% WB	MCDBR	7.8	9.2	12.7	18.1	19.1	16.8	13.4	15.6	14.1	10.0	9.7	9.0
		MCWBR	7.2	7.8	9.7	11.7	11.3	9.8	8.3	10.2	10.0	8.6	8.9	8.0
Clear-Sky Solar Irradiance	taub	0.283	0.298	0.320	0.363	0.386	0.387	0.373	0.377	0.353	0.323	0.280	0.245	
	taud	2.130	2.297	2.273	2.194	2.222	2.305	2.391	2.383	2.412	2.346	2.151	2.058	
	Ebn at Noon	155	227	258	264	266	268	270	259	245	213	154	123	
	Edh at Noon	15	21	29	38	40	37	34	31	26	20	14	10	
All-Sky Solar Radiation	RadAvg	115	356	812	1334	1681	1752	1574	1236	803	406	161	68	
	RadStd	13	47	76	96	126	80	78	117	102	47	21	8	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	-401	N/A	N/A
(47)	Regional (2 neighbors)	N/A	+2.65	+2.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page